Research Bulletin

University of Greater Manchester | Academic Centre Ras al Khaimah

Issue 1 | July 2026 | Director's Cut

School of Engineering

Engineering is the art of solving yesterday's problem just in time for tomorrow's experiment.

Smart Multi-Sensor Fire Detection

One-line signal

This project shows that a simple, well-tuned multi-sensor alarm can become much sharper and more trustworthy when smoke, temperature, and flame evidence are allowed to confirm each other instead of acting alone.

Why this feels promising

There is a strong practical idea underneath the simulation: do not ask one sensor to solve the entire fire problem. Smoke can be noisy, temperature can be slow, and flame can be situational. Together, they create a much fairer decision.

The core build

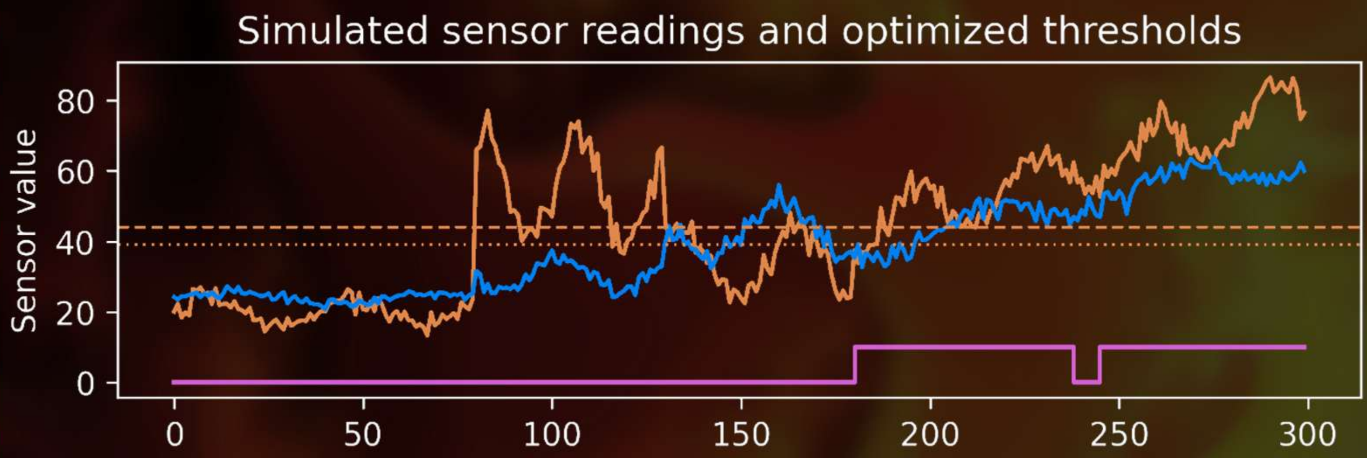
MATLAB was used to simulate scenarios and optimize thresholds.

Simulink was used to implement the final decision logic.

Inputs: smoke level, temperature, and flame presence.

Fire rule: smoke and temperature both cross threshold, or flame appears together with elevated smoke.

Alarm latch: once fire is detected, the alarm stays on until reset.



What stands out

Single-sensor logic was noticeably weaker.

A simple OR rule was fast but far too alarm-happy.

Threshold optimization made the multi-sensor logic more defensible.

The latch solved an important safety behaviour issue by preventing brief alarm dropouts during a real fire sequence.

Measure	Final optimized + latched model
Accuracy	98.0%
False alarm rate	0.0%
Missed detection rate	5.0%
Response time	6 s

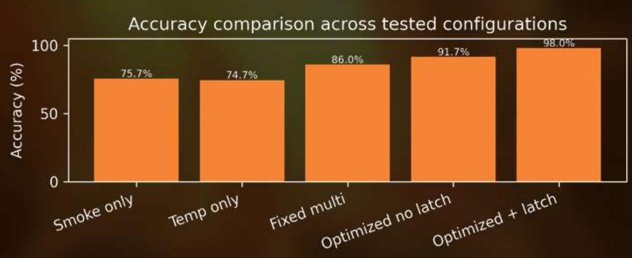
This is the kind of project that could grow well. With real sensors, controlled lab trials, and nuisance-source testing, it has a clear path from a careful simulation exercise to a credible embedded fire-detection prototype.

Where the paper is strongest

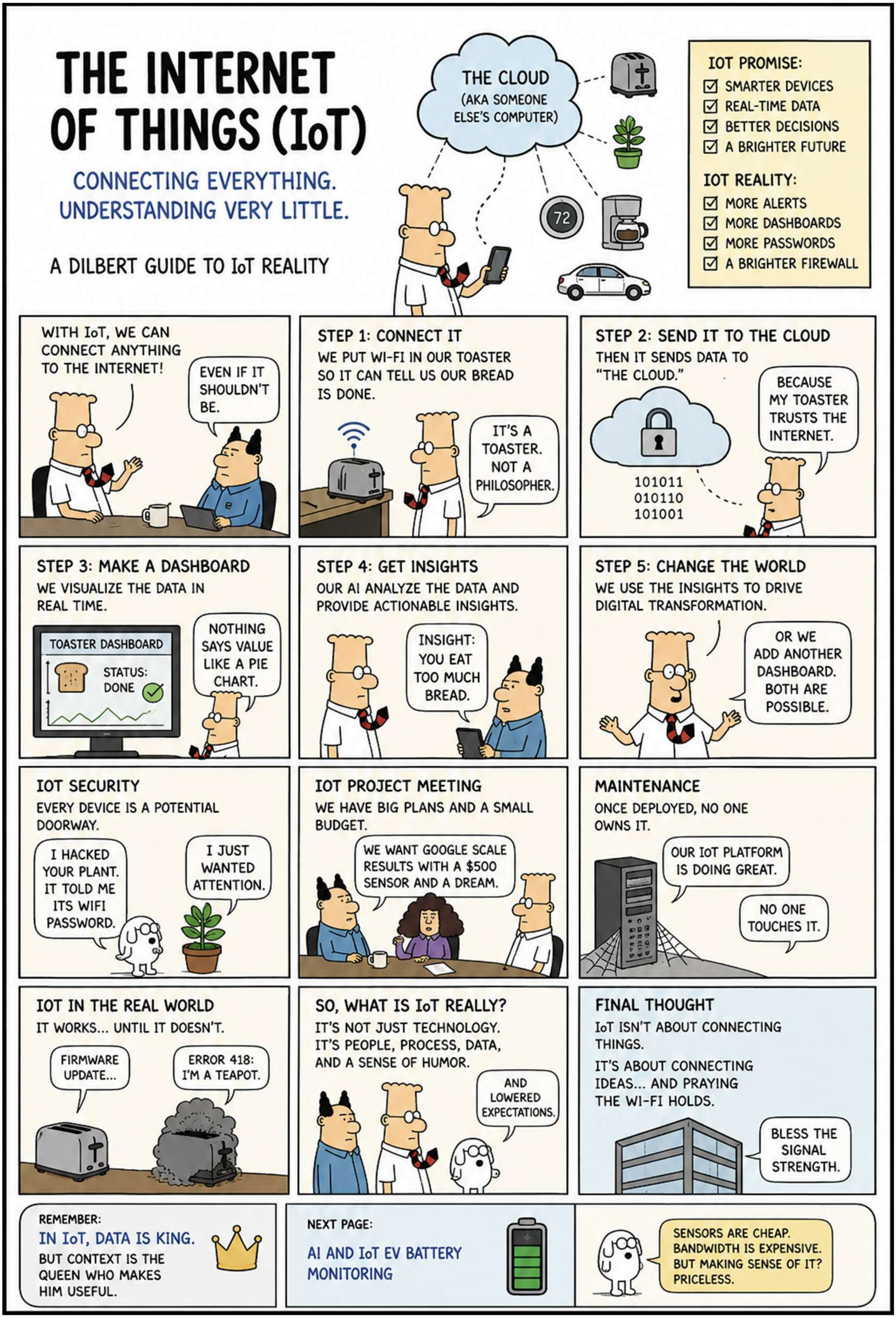
The strongest claim is not that this is a finished commercial detector. It is that transparent logic, tuned carefully, can close part of the gap between crude fixed-threshold systems and heavier AI classifiers.

Honest boundary

The signals were simulated, not measured in a live fire environment. So the result should be read as a strong algorithm-validation study, with real hope for hardware testing later, not as proof of field-ready detector performance.



Deone Zarate



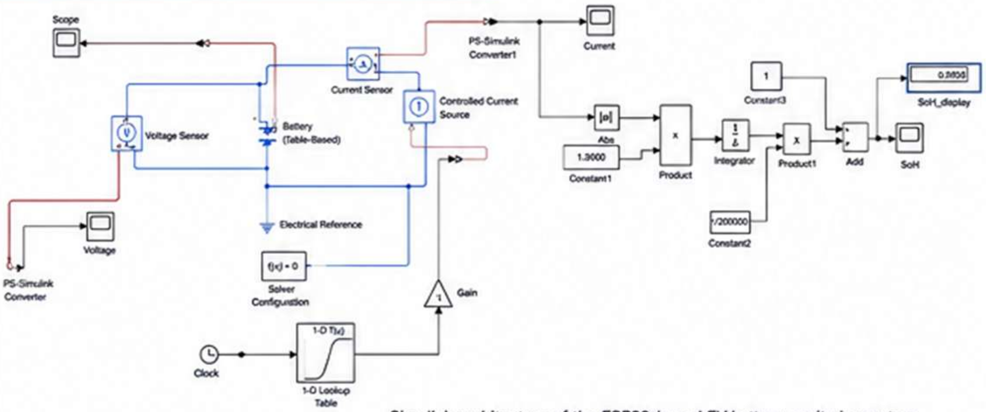
THE INTERNET OF THINGS (IoT)
CONNECTING EVERYTHING. UNDERSTANDING VERY LITTLE.
A DILBERT GUIDE TO IoT REALITY
THE CLOUD (AKA SOMEONE ELSE'S COMPUTER)
72
IOT PROMISE:
☑ SMARTER DEVICES
☑ REAL-TIME DATA
☑ BETTER DECISIONS
☑ A BRIGHTER FUTURE
IOT REALITY:
☑ MORE ALERTS
☑ MORE DASHBOARDS
☑ MORE PASSWORDS
☑ A BRIGHTER FIREWALL
WITH IoT, WE CAN CONNECT ANYTHING TO THE INTERNET!
EVEN IF IT SHOULDN'T BE.
STEP 1: CONNECT IT
WE PUT WI-FI IN OUR TOASTER SO IT CAN TELL US OUR BREAD IS DONE.
IT'S A TOASTER. NOT A PHILOSOPHER.
STEP 2: SEND IT TO THE CLOUD
THEN IT SENDS DATA TO "THE CLOUD."
101011 010110 101001
BECAUSE MY TOASTER TRUSTS THE INTERNET.
STEP 3: MAKE A DASHBOARD
WE VISUALIZE THE DATA IN REAL TIME.
TOASTER DASHBOARD
STATUS: DONE
NOTHING SAYS VALUE LIKE A PIE CHART.
STEP 4: GET INSIGHTS
OUR AI ANALYZE THE DATA AND PROVIDE ACTIONABLE INSIGHTS.
INSIGHT: YOU EAT TOO MUCH BREAD.
STEP 5: CHANGE THE WORLD
WE USE THE INSIGHTS TO DRIVE DIGITAL TRANSFORMATION.
OR WE ADD ANOTHER DASHBOARD. BOTH ARE POSSIBLE.
IOT SECURITY
EVERY DEVICE IS A POTENTIAL DOORWAY.
I HACKED YOUR PLANT. IT TOLD ME ITS WIFI PASSWORD.
I JUST WANTED ATTENTION.
IOT PROJECT MEETING
WE HAVE BIG PLANS AND A SMALL BUDGET.
WE WANT GOOGLE SCALE RESULTS WITH A $500 SENSOR AND A DREAM.
MAINTENANCE
ONCE DEPLOYED, NO ONE OWNS IT.
OUR IoT PLATFORM IS DOING GREAT.
NO ONE TOUCHES IT.
IOT IN THE REAL WORLD
IT WORKS... UNTIL IT DOESN'T.
FIRMWARE UPDATE...
ERROR 418: I'M A TEAPOT.
SO, WHAT IS IoT REALLY?
IT'S NOT JUST TECHNOLOGY. IT'S PEOPLE, PROCESS, DATA, AND A SENSE OF HUMOR.
AND LOWERED EXPECTATIONS.
FINAL THOUGHT
IoT ISN'T ABOUT CONNECTING THINGS.
IT'S ABOUT CONNECTING IDEAS... AND PRAYING THE WI-FI HOLDS.
BLESS THE SIGNAL STRENGTH.
REMEMBER:
IN IoT, DATA IS KING.
BUT CONTEXT IS THE QUEEN WHO MAKES HIM USEFUL.
NEXT PAGE:
AI AND IoT EV BATTERY MONITORING
SENSORS ARE CHEAP. BANDWIDTH IS EXPENSIVE. BUT MAKING SENSE OF IT? PRICELESS.

AI AND IOT EV BATTERY MONITORING

From Sensing to Cloud to State-of-Health Prediction

> "A clean proof-of-concept that makes battery data visible, connected, and analytically useful."

ONE-LINE SIGNAL ARCHITECTURE



Simulink architecture of the *ESP32-based EV battery monitoring system* from sensing to SOH estimation.

ESP32 Sensing (Wokwi) → Wi-Fi Transmission → ThingSpeak Cloud → MATLAB/Simulink Analytics → SOH Prediction & Insights

THE CORE WORKFLOW

1. Wokwi was used to simulate an ESP32-based sensing setup.
2. Voltage, current, and temperature were uploaded through Wi-Fi to ThingSpeak.
3. SOC was derived from the monitored signals.
4. MATLAB and Simulink were used for SOH-related processing and prediction.
5. The study compared healthy, medium, poor, and critical battery-condition scenarios.

WHY THIS FEELS PROMISING

The project does not overcomplicate the architecture. It uses an ESP32, ThingSpeak, and MATLAB in a sequence that is easy to understand, easy to extend, and genuinely relevant to future battery-supervision ideas.

KEY MESSAGE IN PLAIN LANGUAGE

The paper is at its best when read as an integration success. It shows that battery telemetry, cloud dashboards, and a prediction stage can all live in one coherent workflow without expensive hardware.

WHAT TO CELEBRATE CAREFULLY

The hopeful part is not that battery prognostics has been solved. The hopeful part is that the pipeline works end to end. That matters because many student projects stop at sensing or stop at dashboards. This one pushes through to analytics.

RESULTS WORTH KEEPING

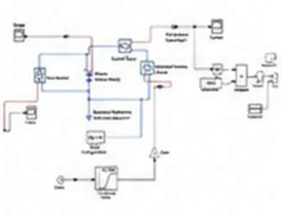
- The dashboard clearly separated healthier and degraded operating states.
- Rising temperature and falling SOC created an intuitive warning pattern.
- A regression-based SOH prediction stage ran successfully on exported data.
- Reported RMSE was low, but the prediction range and dataset were both very small.

HONEST BOUNDARY

The inputs were simulated, and the SOH labels came from a simplified cycle-count rule rather than experimental battery aging data. That means the prediction result is best treated as feasibility evidence, not validated prognostics performance.

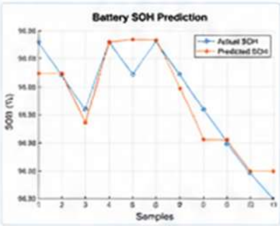
VISUAL SPREAD – FROM DATA TO INSIGHT

1. ESP32 SIMULATION ARCHITECTURE



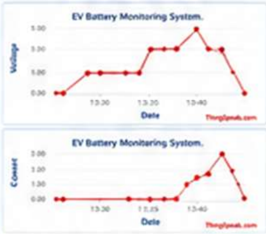
The emulated hardware layer that feeds the rest of the workflow.

2. MATLAB & SIMULINK SOH MODEL



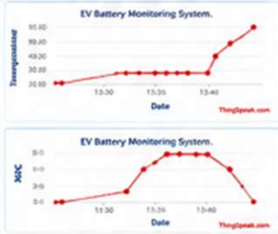
The analytics stage that turns telemetry into health-related interpretation.

3. THINGSPEAK DASHBOARD VOLTAGE & CURRENT



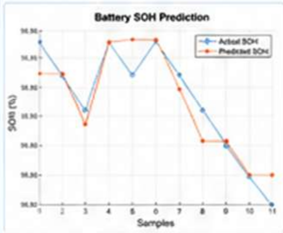
Cloud plots make condition changes visible rather than hidden in raw data streams.

4. THINGSPEAK DASHBOARD TEMPERATURE & SOC



Temperature rise and SOC decline provide the most readable operating story.

5. ACTUAL vs PREDICTED SOH



A useful first prediction view, even if the dataset is still too small for strong claims.

BULLETIN TAKEAWAY

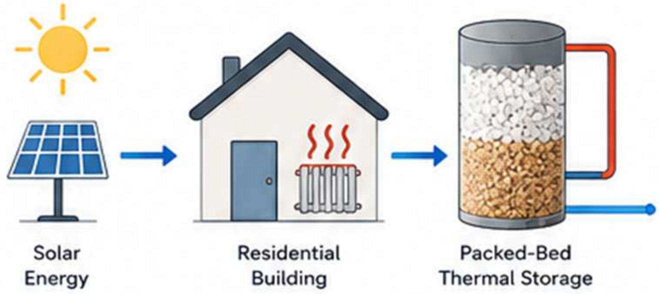
This project opens a practical door. With real battery data, stronger labels, and larger operating histories, the same architecture could evolve from a smart student prototype into a serious small-scale battery-health research platform.

SUSTAINABLE MOBILITY | CLOUD CONNECTIVITY | AI & ANALYTICS FOR HEALTH | SMARTER BATTERIES SAFER FUTURES | Scan for more details

BRINE-DERIVED SALT FOR PACKED-BED THERMAL ENERGY STORAGE

A Feasibility Study of Brine Waste as an Alternative Thermal Storage Medium

Turning desalination waste into practical residential thermal storage.



ONE-LINE SIGNAL

This study investigates whether a brine-derived chloride-rich salt proxy can replace sand in a residential packed-bed thermal energy storage (TES) system while maintaining indoor thermal comfort.

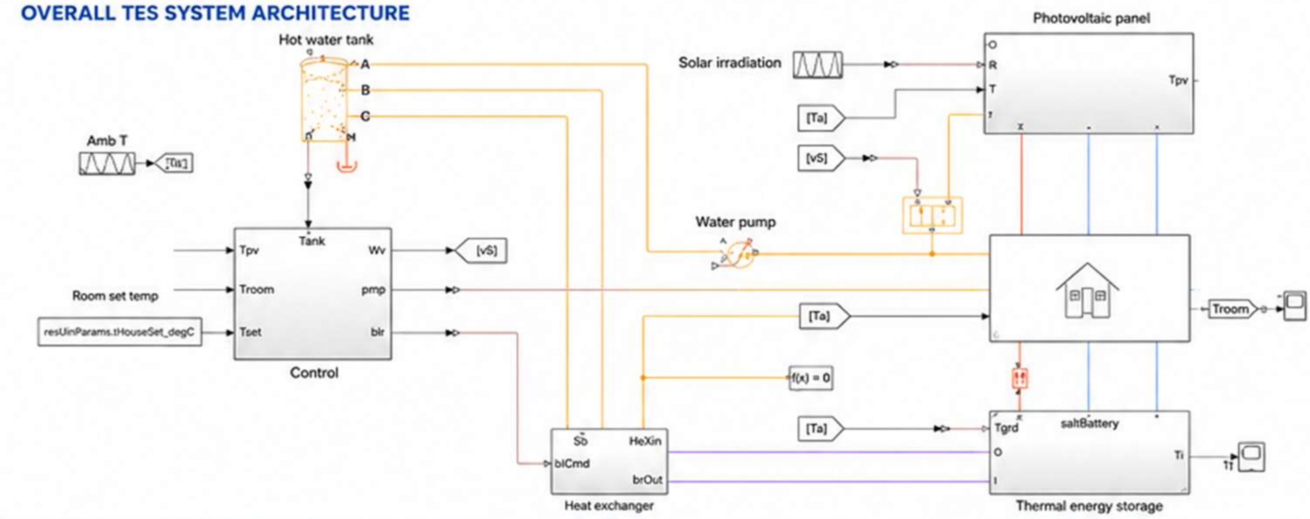
WHY THIS MATTERS

Waste Recovery
Reuse desalination by-products instead of disposal.

Residential Energy
Investigate low-cost seasonal heat storage.

Sustainable Materials
Explore alternative storage media with future circular-economy potential.

OVERALL TES SYSTEM ARCHITECTURE



COMPARED STORAGE MEDIA

	Sand	Salt Proxy
Type	Baseline	Alternative
Description	Well-established granular material	Chloride-rich granular material
Behavior	Higher charging temperature	Lower charging temperature

ASSUMED SALT PROPERTIES

- Density: 1850 kg/m³
- Thermal Conductivity: 1.2 W/m-K
- Specific Heat: 1500 J/kg-K

SIMULATION CONDITIONS

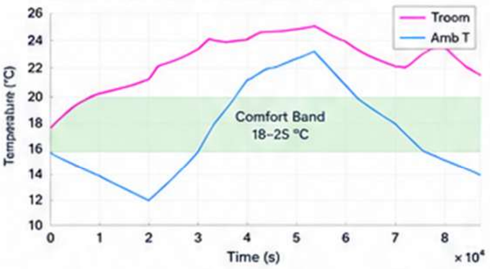
- Same residential TES model
- Same geometry
- Same controls
- Same 24-hour simulation

Only storage medium properties changed.

KEY FINDINGS

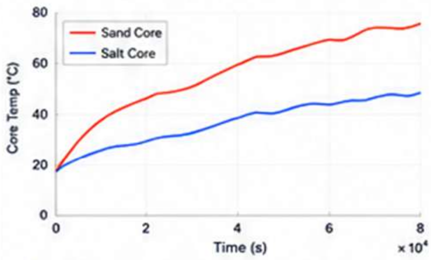
- Sand storage reached approximately **74°C.**
- Salt proxy stabilized near **49°C.**
- Indoor temperature remained within the **18–25°C** comfort band.
- The replacement changed storage behavior but did not significantly affect room comfort.

ROOM TEMPERATURE RESPONSE



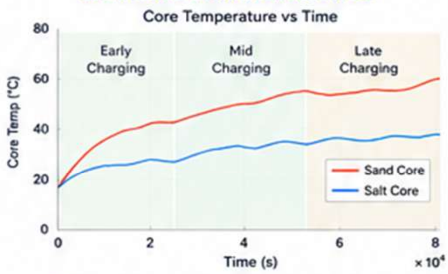
Indoor temperature stayed within the desired comfort range despite lower storage temperature.

CORE TEMPERATURE COMPARISON



Sand storage reached a higher core temperature, while salt proxy charged more gently.

CHARGING CHARACTERISTICS



Different charging behavior: sand heats faster; salt proxy shows a slower, steadier rise.

ENGINEERING INSIGHT

> Higher storage temperature does not necessarily translate into better residential performance. Controlled, comfort-preserving heat delivery is often more valuable than maximizing storage temperature.

STUDY BOUNDARY

This work is an early-stage simulation using a proxy material.

Not included:

- Experimental material characterization
- Corrosion analysis
- Moisture behaviour
- Cycling durability
- Brine recovery energy

Therefore, the results demonstrate feasibility, not deployment readiness.

BULLETIN TAKEAWAY

This study provides encouraging evidence that brine-derived salt deserves further laboratory investigation as a low-cost thermal storage medium. Future work should validate material properties experimentally and evaluate long-term cycling performance.

RESEARCH BULLETIN

OPTIMIZING SOLAR HARVEST

Dust-Aware PV Tilt Optimization for Rooftop Solar in Abu Dhabi

ONE-LINE SIGNAL

This study demonstrates that carefully selecting the tilt angle of rooftop photovoltaic panels—while accounting for desert dust and high operating temperatures—can substantially improve annual energy production, financial returns, and carbon savings for commercial buildings in Abu Dhabi.

WHY THIS FEELS PROMISING

Instead of assuming a standard tilt, this work uses PVsyst simulations with local weather and soiling conditions to determine the best orientation specifically for Abu Dhabi.

The result is a methodology that designers can reuse for similar commercial rooftop installations across the UAE.

ENGINEERING CHALLENGE

Extreme Solar Resource
Abu Dhabi has excellent solar irradiation but also very high module temperatures.

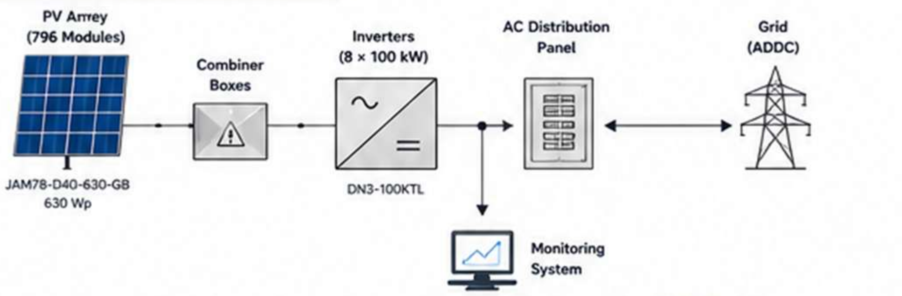
Dust & Soiling
Dust accumulation reduces PV output unless the system is cleaned or installed at favourable tilt angles.

Commercial Retrofit
Older glazed office buildings have high cooling loads, making rooftop PV an attractive retrofit option.

CORE WORKFLOW

1. Site Survey → 2. PVsyst System Design (DN3-100KTL) → 3. Tilt & Azimuth Scan → 4. 8760-hour Simulation → 5. Financial Analysis → 6. Carbon Assessment (CO_2)

PVSYST SYSTEM ARCHITECTURE



Simulation workflow used to optimize rooftop PV orientation for ASCORP Holding.

SIMULATION SETTINGS

- **Location:** Abu Dhabi (24.3343°N, 54.4894°E)
- **Weather Data:** Meteonorm 9.0
- **Soiling Losses:** 3–5% (quarterly cleaning)
- **Temperature Model:** NOCT with local wind
- **Degradation:** 1.0%/year (linear)
- **Transposition Model:** Perez diffuse
- **Simulation:** 8760 hourly steps (1 year)

OPTIMAL ORIENTATION

Tilt Angle
26.3°

Azimuth
0° (South)

Global on Collector
2,285 kWh/m²

Transposition Factor
1.08

This orientation delivered the highest annual solar collection.

ENERGY PERFORMANCE

Annual Energy
893 MWh/year

Performance Ratio (PR)
0.783

Capacity Factor
20.4%

Specific Production
1,783 kWh/kW

Performance ratio exceeds regional benchmarks (0.75–0.80).

FINANCIAL PERFORMANCE

LCOE
0.0351 USD/kWh

Payback Period
5.2 years

25-year Net Profit
$2.25 Million

Approximately 65% cheaper than the commercial grid tariff.

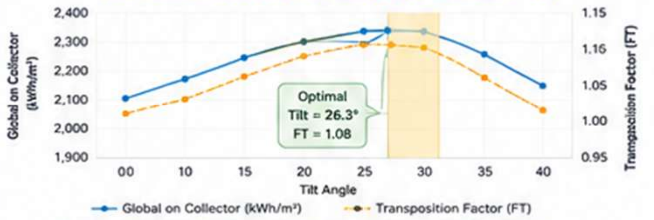
ENVIRONMENTAL IMPACT

Annual CO_2 Reduction
438.7 tonnes

Lifetime CO_2 Reduction (25 years)
13,161 tonnes

Equivalent to removing ~95 passenger vehicles from the road each year.

TILT PARAMETRIC SCAN RESULTS (AZIMUTH = 0°)



26.3° tilt angle provides the highest annual collection (2,285 kWh/m²), achieving an 8.1% gain compared to horizontal mounting.

COMPARISON WITH LITERATURE BENCHMARKS

Study	System Type	Specific Yield (kWh/kW)	Performance Ratio (PR)	Payback (Years)
This Study	Rooftop (S01 kW)	1,783	0.783	5.2
Hasanain (2024) [1]	Ground-mounted	1,800 – 1,950	–	–
Elnabawi (2021) [4]	ESTIDAMA Pearl 1	–	0.75 – 0.80	–
Masdar NZ1 (2023) [7]	New Construction	1,750	0.78	–

Performance is competitive with literature values despite accounting for temperature derating and 3–5% soiling losses.

KEY TAKEAWAYS

- ✔ Dust-aware optimization identified 26.3° as the optimal tilt angle.
- ✔ 8% increase in energy yield compared to horizontal installation.
- ✔ Strong economics: 5.2-year payback and 65% lower LCOE than grid.
- ✔ High performance ratio (0.783) above regional benchmarks.
- ✔ Significant CO_2 reduction supporting UAE Net Zero 2050 target.

STUDY BOUNDARY

- Simulation-based design (not yet operational).
- Single-year weather data (Meteonorm).
- Simplified shading analysis.
- Zero discount rate assumed.
- Quarterly cleaning schedule assumed.
- No field validation yet.

Results demonstrate potential and feasibility, not operational performance.

BULLETIN TAKEAWAY

This study shows that location-specific PV design matters. By incorporating local dust behavior, high temperatures, and real-world conditions into tilt and azimuth selection, commercial rooftop systems can deliver higher energy, stronger financial returns, and meaningful environmental benefits.

ACKNOWLEDGMENT

The authors thank Ms. Anu Lonappan and Dr. Prasanthi Achikkulath for their supervision and support, and the University of Greater Manchester Ras Al Khaimah Academic Centre for providing the academic resources required for this project.